

Project Summary



CareerFlow AI

From profile to job readiness

A web app that reads a person's CV and guides them, step by step, toward a suitable job — in **any field**, not only IT.

Version 2.4.0 · October 2026

8	11	33	40	7
core features	API endpoints	roles in 18 fields	automated tests	small packages

1. What it does

Feature	Description
Resume Analyzer	Upload a PDF CV; it extracts name, skills, experience, education, projects, certifications and location into an editable profile. You can also enter the profile manually.
Recommended Roles	Suggests about 6 roles that fit <i>your</i> CV in any field, each with a 0–100 match score and a “Why it fits” line. Suggest different roles gives a fresh set without repeats; a preference box (e.g. “finance”, “remote”) steers results; Set as my target accepts any job title you type.
Skill Gap Analysis	For the target role: skills you already have, required skills that are missing (learn first), and nice-to-have skills.
Learning Plan	A 7-day plan aimed at your missing skills, with official learning links, tick-off progress and a Regenerate button.
Interview Simulator	Questions for your target role at Easy, Medium or Hard level, rotating technical, behavioral and scenario types. Scored feedback (correctness, completeness, clarity) and a saved practice history.
CV Builder	4 templates (Modern, Classic, Minimal, Professional) and 6 accent colours. Prefilled from your profile, AI-written summary, a missing-details check with “Add now” buttons, and A4 PDF download.
Career Coach chat	A header button opens a chat that gives short (about 90 words), personal advice using your profile, target role and skill gaps.
Dashboard	Readiness score, matched roles, skill gaps and plan progress at a glance, with light and dark mode.

Everything follows the role you select. Changing the target role updates the skill gap, the learning plan and the interview questions, so the whole journey stays aligned with one career path.

2. User journey

1	2	3	4	5	6	7
Upload CV	Review profile	Pick a role	See skill gap	Follow 7-day plan	Practise interview	Build CV

The Career Coach is available on every screen for questions along the way.

3. How it's built

Part	Technology
Frontend	Plain HTML, CSS and JavaScript — no framework and no build step
Backend	Python FastAPI, 11 API endpoints plus a health check
AI	Groq (openai/gpt-oss-120b) first, Gemini as backup — called directly over HTTPS, no vendor SDKs
Offline mode	Every feature has a built-in, rule-based fallback, so the app keeps working if the AI is unavailable
Search	In-memory TF-IDF matching of profiles, roles and skills
Data	JSON files: 33 roles across 18 fields, 32 learning resources, skills, careers and interview guidance
Storage	No database. Each user's data stays in their own browser; uploaded CVs are never stored on the server
Footprint	7 small Python packages: fastapi, uvicorn, python-multipart, pydantic, python-dotenv, httpx, pypdf

Security for public use

- Per-visitor rate limits (30 AI requests and 120 other requests per minute)
- Upload size limit (5 MB) and size limits on every input field
- Browser security headers, including a strict Content-Security-Policy
- Defences against instructions hidden inside CVs or chat messages
- API keys stay on the server and are never sent to the browser

Quality

40 automated tests (matching, skills, offline logic, AI parsing, all API endpoints, a real PDF upload) and a GitHub Actions check that runs the tests and a Docker build on every push.

4. API endpoints

Method & path	Purpose
GET /api/health	Status, AI configuration, search index size
POST /api/resume/analyze	PDF upload → structured profile
POST /api/resume/improve	Resume suggestions for the target role
POST /api/jobs/recommend	Roles suggested from the CV (supports exclude and preference)
POST /api/jobs/custom	Build a target role from a typed job title
POST /api/skills/gap	Have / missing / nice-to-have skills
POST /api/learning-plan	7-day plan with learning resources
POST /api/interview/start	Next question (easy, medium or hard)
POST /api/interview/evaluate	Scores 0–10 with short feedback
POST /api/chat	Career Coach reply (kept to about 90 words)
POST /api/cv/summary	Draft a CV summary from the profile

5. Project layout

```
careerflow-ai/  
  backend/app/      main.py, config.py, schemas.py, check_ai.py  
    api/routes.py   all endpoints  
    core/security.py rate limits, security headers  
    services/       AI, matching, search, career logic, skills, PDF  
    data/           roles, skills, learning, careers, interviews (JSON)  
  backend/tests/    40 tests + sample CV  
  frontend/         index.html, script.js, style.css, theme.js, config.js, logo  
  docs/DEPLOYMENT.md  
  Dockerfile, render.yaml, run.bat, .env.example, README.md
```

6. Running and hosting

- **Locally:** double-click `run.bat`. It sets up the environment, checks the AI keys, starts the server and opens `http://127.0.0.1:8000`.
- **Online:** not deployed yet. The project is ready for free hosting on **Render** (render.yaml blueprint included) or **Hugging Face Spaces**, both using the same Dockerfile. Step-by-step instructions are in `docs/DEPLOYMENT.md`.
- **Configuration:** `GROQ_API_KEY` and/or `GEMINI_API_KEY` in a `.env` file locally, or in the host's secrets panel online.

7. Known notes

- Groq works on the development machine; Gemini (the backup) is currently blocked by that network with an SSL error. Nothing depends on it.
- Suggested roles are career directions, not live job vacancies. Match scores and interview feedback are guidance, not hiring decisions.
- On Windows, clicking inside the server window pauses it (Select mode); press Esc to resume.
- Render's free plan sleeps after 15 minutes without visitors and takes about a minute to wake up.